## Chessboard Problems

**Problem 1.** A knight is modified so that it moves $p$ fields horizontally or vertically and $q$ fields in the perpendicular direction (where $p, q \neq 0$). It is placed on an infinite chessboard. If the knight returns to the initial field after $n$ moves, show that $n$ must be even.

**Problem 2.** $X$ and $Y$ play a game on an $m \times n$ chessboard, where $m$ and $n$ are positive integers. Initially, the top left-hand corner cell is marked, and there is a rook in this marked cell. Starting from $X$, the two players take turn moving the rook to another cell in the same row or the same column such that all cells in between are unmarked (including the new cell that it stays in). Then all these cells in between are marked. The player who cannot move will lose the game. Find all $m$ and $n$ such that $X$ has a winning strategy.

**Problem 3.** Originally, every square of an $8 \times 8$ chessboard contains a rook. One by one, rooks which attack an odd number of others are removed. Find the maximum number of rooks that can be removed. (A rook attacks another rook if they are on the same row or column and there are no other rooks between them.)

**Problem 4.** A chess knight has injured his leg and is limping. He alternates between a normal move and a short move where he moves to any diagonally neighbouring cell. The limping knight moves on a $5 \times 6$ chessboard starting with a normal move. What is the largest number of moves he can make if he is starting from a cell of his own choice and is not allowed to visit any cell (including the initial cell) more than once?

**Problem 5.** Let $n \geqslant 2$ be an integer. Consider an $n \times n$ chessboard with the usual chessboard colouring. A move consists of choosing a $1 \times 1$ square and switching the colour of all squares in its row and column (including the chosen square itself). For which $n$ is it possible to get a monochrome chessboard after a finite sequence of moves?

**Problem 6.** Given an $n \times n$ chessboard, where $n \geqslant 4$ and $p = n + 1$ is a prime number. A set of $n$ unit squares is called *tactical* if after putting down queens on these squares, no two queens are attacking each other. Prove that there exists a partition of the chessboard into $n - 2$ tactical sets, not containing squares on the two diagonals. (Queens are allowed to move horizontally, vertically and diagonally.)

**Problem 7.** Some cells of a $10 \times 10$ chessboard are coloured blue. Determine the greatest value of $n$ for which it is possible to colour $n$ chessboard cells blue such that there is no $3 \times 2$ or $2 \times 3$ rectangle which is entirely blue.

**Problem 8.** Some cells of a $10 \times 10$ chessboard are coloured blue. A set of six cells is called *gremista* when the cells are the intersection of three rows and two columns, or two rows and three columns, and are painted blue. Determine the greatest value of $n$ for which it is possible to colour $n$ chessboard cells blue such that there is not a gremista set.